



CUSTOMER SPECIFICATION DOCUMENT

CSD-1322

MightySat Bluetooth Communication Protocol Specification

Revision: B
CO-067126

APPROVALS

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1 PURPOSE

Masimo MightySat provides data communications facilities through asynchronous Bluetooth Low Energy. This document describes the Masimo MightySat communication protocol that can be used to communicate with Masimo MightySat instruments through Bluetooth Low Energy as underlying transfer layer in order to make available parametric data to external devices. This document will enable the designers and implementers of such a connected system to establish the communication and software protocol necessary in order to correctly communicate with the Masimo MightySat.

2 OVERVIEW

This communication protocol is intended to only provide data output. The communication protocol is referred to as the “MightySat” communication protocol.

3 PROTOCOL SPECIFICATION

This protocol specification describes the data format as Little Endian.

3.1 Advertisement Packet

During scanning of devices, it is recommended to filter for MightySat devices by using the Manufacturer Specific Data block. Devices manufactured by Masimo will use the Company ID of 0x0243.

Filtering/Scanning by Service UUID is not reliable because the Service UUID field availability varies from device to device depending on whether or not device supports the standard BLE Pulse Oximeter Profile (PLXP).

- This means there are two expected formats for the Advertisement packet described below.

3.1.1 Data Format – PXL P Support

Description		Value (s)
Flags (3 bytes)	Len	0x02
	Type	0x01
	Data	0x06
Service UUID (4 bytes)	Len	0x03
	Type	0x02
	Data (2 bytes)	0x1822
Device Name (11 bytes)	Len	0x0A
	Type	0x09
	Data (9 bytes)	“MightySat”
Manufacturer Specific Data (11 bytes)	Len	0x0A
	Type	0xFF
	Company ID (2 bytes)	0x0243
	Structure Version	0x01
	Product Type	0x01
	Product Variant	0x00 = Consumer Version 0x01 = RX Version
	Serial Number in Hex (4 bytes)	0XXXXXXXX (uint32)

3.1.1 Data Format – No PXL P Support

When the devices does not support the PXL P profile, no Service UUID field is advertised.

Description		Value (s)
Flags (3 bytes)	Len	0x02
	Type	0x01
	Data	0x06
Device Name (11 bytes)	Len	0x0A
	Type	0x09
	Data (9 bytes)	“MightySat”
Manufacturer Specific Data (11 bytes)	Len	0x0A
	Type	0xFF
	Company ID (2 bytes)	0x0243
	Structure Version	0x01
	Product Type	0x01
	Product Variant	0x00 = Consumer Version 0x01 = RX Version
	Serial Number in Hex (4 bytes)	0XXXXXXXX (uint32)

3.1.2 Product Variant

The MightySat Communication Protocol supports two product variants which is reflected in the Manufacturer Specific Data block:

Product Variant (1 byte)	Description
0x00	The device is <u>consumer version</u> supporting three parameters: SpO2, PR, PI
0x01	The device is the <u>RX version</u> supporting five parameters: SpO2, PR, PI, PVi, RRp

3.2 MightySat GATT Service

MightySat implements a proprietary protocol on a custom GATT service.

3.2.1 Service & Characteristics

MightySat broadcasts its 128-bit UUID as a way to identify its service to the Host. This UUID is uniform for all MightySat units. Its equivalent 16-bit UUID is identified starting from the offset byte 2 (based -0 index) of the long 128-bit UUID.

<i>MightySat 128-bit Service UUID</i>	<i>54c21000-a720-4b4f-11e4-9fe20002a5d5</i>
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This service provides two characteristics: the transmitting characteristic, and the receiving characteristic, respectively. These two available characteristics from the MightySat are indicated below:

<i>16-bit Service ID</i>	<i>Property</i>	<i>Notes</i>
0x1001 (<i>RX</i>)	Write-Without-Response	Hosts can send messages (i.e. commands) to the MightySat through this characteristic.
0x1002 (<i>TX</i>)	Notify Only	Hosts can receive messages (i.e. measurements, responses) from the MightySat through this characteristic.

3.3 Message Format

Messages transporting between the MightySat and the Host are streams of bytes. Each message starts with a SOM sentinel. Message integrity is ensured by the CRC and the length information.

<i>SOM</i>	<i>LEN (1 byte)</i>	<i>PAYLOAD (N bytes)</i>	<i>CRC (1 byte)</i>
0x77	N	$d1, d2 \dots dN$	$crc(d1, d2 \dots dN)$

- SOM is the sentinel Start-of-Message character.
- LEN is equal to the length of the enveloped message and the CRC but does not include LEN itself
 - $LEN = SIZEOF(PAYLOAD + CRC)$
 - Valid LEN values are 1-255.
- CRC is the value computed using CRC-8-CCITT for bytes after LEN and before CRC.
 - The CRC-8-CCITT algorithm is a standard 8-bit cyclic redundancy check using the following configurations:
 - polynomial $P(x) = x^8 + x^2 + x + 1$
 - initial CRC seed of 0.
- 16 and 32 bit data values are unpacked into bytes LSB (Least Significant Byte) first.

3.4 Commands & Responses

Commands are messages can be sent from the Host to MightySat via the RX characteristic (0x1001). Responses are messages that are sent from MightySat to the Host via the TX characteristic (0x1002).

<i>Command</i>	<i>ID</i>	<i>Reference</i>
Get Device Information	0x01	See 3.4.1
Set Clock	0x02	See 3.4.2
Configure Streaming Data	0x03	See 3.4.3
Get Trend Record	0x06	See 3.4.4
Clear Trend Records	0x07	See 3.4.5

Table 1: Commands

Commands that support NACK responses will contain Error Codes per the following table:

<i>Code</i>	<i>Name</i>
0	None
1	Record Not Founded
2	Command Failed
3	Not Supported
4	Record Corrupted
5	Bad CRC

Table 2: Error Codes

3.4.1 Get Device Information

In order to retrieve details about the device and its configuration, this **command** is sent from the Host to MightySat:

PAYLOAD (1 byte)	
ID (1 byte)	Data (0 bytes)
0x01	N/A

Table 3: Get Device Information Command

Example of a command sent to MightySat: [0x77, 0x02, 0x01, 0x07]

This **response** is sent from MightySat to the Host:

PAYLOAD (19 bytes)	Size	Description	Value (s)
	1 byte	Get Device Information Command ID	0x01
	2 bytes	SW Version	(e.g. 0x1000)
	2 bytes	Available Parameters	See Table 4: Parameters Bitmask
	1 byte	Available Waveforms	See Table 4: Waveforms Bitmask
	2 bytes	Trend version	(e.g. 0x1000)
	2 bytes	Trended Parameters	See Table 3: Parameters Bitmask
	1 byte	Number of trend record (uint8)	[0-100]
	4 bytes	Oldest Trend Session Id	(e.g. 0x00000000)
	4 bytes	Current Trend Session Id	(e.g. 0x00000063)

Table 4: Get Device Information Response

Bit	Parameter
0	SpO2
1	PR
2	PVI
3	PI
4	RRp
5 – 15	Reserved

Table 5: Parameters Bitmask

Bit	Parameter
0	Pleth
1	SIQ
2 – 7	Reserved

Table 6: Waveforms Bitmask

3.4.2 Set Clock

In order to set the reference time used to timestamp trend record data that can be retrieved later, this **command** is sent from the Host to MightySat:

<i>PAYLOAD (5 bytes)</i>	
<i>ID (1 byte)</i>	<i>Parameters</i>
0x02	Seconds since Unix Epoch (4 bytes, uint32)

Table 7: Set Clock Command

Example of a command sent to MightySat: [0x77, 0x06, 0x02, 0x4D, 0x74, 0xEE, 0x5F, 0x62]

If the command succeeds, this **ACK response** is sent from MightySat to the Host:

<i>PAYLOAD (2 bytes)</i>	
<i>ID (1 byte)</i>	<i>Data (1 byte)</i>
0xFE	0x02

If the command fails, this **NACK response** is sent from MightySat to Host:

<i>PAYLOAD (3 Bytes)</i>		
<i>ID (1 byte)</i>	<i>Data (2 bytes)</i>	
0xFF	0x02	Error Code (1 byte, uint8)
		See Table 2: Error Codes

3.4.3 Configure Streaming Data

In order to configure which parameters and waveforms shall be streaming for the MightySat, this **command** is sent from the Host to MightySat:

PAYLOAD (4 bytes)		
ID (1 byte)	Data (3 bytes)	
0x03	Parameters Bitmask (2 bytes) See Table 5	Waveforms Bitmask (1 byte) See Table 6

Table 2: Configure Streaming Data Command

Example of a command sent to MightySat: [0x77, 0x05, 0x03, 0x1F, 0x00, 0x03, 0xD6]

If the command is successful, this **ACK response** is sent from the MightySat to the Host:

PAYLOAD (2 bytes)	
ID (1 byte)	Data (1 byte)
0xFE	0x03

- After a successful command and the ACK response, the device will start periodic transfer of waveforms and/or parameters depending on the bitmask. (1 = enable, 0 = disable)

If the command fails, this **NACK response** is sent from MightySat to Host:

PAYLOAD (3 bytes)		
ID (1 byte)	Data (2 bytes)	
0xFF	0x03	Error Code (1 byte, uint8) See Table 2: Error Codes

3.4.3.1 Parameter Streaming Response (asynchronous)

The Parameter Streaming **response** is sent from the MightySat to the host every second (i.e. rate @ 1hz).

- There will be one system exception sent for each message.
- To find out what parameters is currently being streamed, see the **Get Device Info** command.
- The order of parameters in each response corresponds to **Parameters Bitmask** order from LSB to MSB.
- All of the parameter values are uint8, except for PI which is a uint16 where the value is multiplied by 100.

<i>PAYLOAD (N Bytes)</i>	<i>Size</i>	<i>Description</i>	<i>Value (s)</i>
	1 byte	Parameter Streaming Response ID	0x05
	4 bytes	System Exceptions Bitmask	See Table 9: System Exceptions Bitmask
	1 byte	Parameter #1 – Parameter Exceptions Bitmask	See Table 10: Parameter Exceptions Bitmask
	1-2 bytes	Parameter #1 – Value	e.g. [0-255]
	...		
	1 byte	Parameter #N – Parameter Exceptions Bitmask	See Table 10: Parameter Exceptions Bitmask
	1-2 bytes	Parameter #N – Value	e.g. [0-255]

Table 8: Parameter Streaming Response

<i>Bit</i>	<i>System Exception Description</i>
0 – 20	<i>Reserved</i>
21	Sensor Off Patient
22	Pulse Search
23	Interference Detected
24	Low Perfusion
25 – 31	<i>Reserved</i>

Table 9: System Exceptions Bitmask

<i>Bit</i>	<i>Parameter</i>
0	Low Confidence
1	<i>Reserved</i>
2	Invalid
3	<i>Reserved</i>
4	Startup State
5 – 7	<i>Reserved</i>

Table 10: Parameter Exceptions Bitmask

3.4.3.2 Waveform Streaming Response (asynchronous)

The Waveform Streaming **response** is sent from the MightySat to the host whenever buffered samples are available:

- The Ordinal is the count of all previous samples sent (excluding the samples of the current packet). The count will wraparound. This field is important to identify if packets are missing.
 - e.g. n ordinal value of 1 means a single pair of Pleth and SIQ samples have been sent previously.
- The response can contain 1 to N samples.
- The sampling period is 32 ms (i.e. rate @ 31.25hz) for each sample. This is important for waveform plotting.
- Each sample contains 2 values: Pleth, and SIQ.

PAYLOAD (N Bytes)	Size	Description	Value (s)
	1 byte	Waveform Streaming Response ID	0x04
	1 byte	Ordinal (uint8)	[0 – 255]
	1 byte	Sample #1 - Pleth value (int8)	[-128 to 127]
	1 byte	Sample #1 - SIQ value	See Table 12: SIQ Value
	...		
	1 byte	Sample #N - Pleth value (int8)	[-128 to 127]
	1 byte	Sample #N - SIQ value	See Table 12: SIQ Value

Table 11: Waveform Stream Response

Bit	Description
0 – 6	Unsigned integer value [0 – 100]
7	Invalid

Table 12: SIQ Value

3.4.4 Get Trend Record

In order to retrieve trend records from MightySat, this **command** is sent from the Host to MightySat:

- Developers should not request next record before receiving response (or NAK) from previous request.
- The Trend Session ID is incremented every time the user removes the MightySat from the finger and the displayed values freeze.
- PI is not save as a trend record, even though it may be configured for streaming.
- When the value is invalid, it is set to 255.

PAYLOAD (4 bytes)	
ID (1 byte)	Data (4 bytes)
0x06	Trend Session ID (uint32)

Table 13: Get Trend Record Command

Example of a command sent to MightySat: [0x77, 0x06, 0x06, 0x9B, 0x00, 0x00, 0x00, 0x97]

If the command is successful, this **response** is sent from the MightySat to the Host:

PAYLOAD (N Bytes)	Size	Description	Value (s)
	1 byte	Get Trend Record command ID	0x06
	4 bytes	Trend Session Id	e.g. 0x00000000
	4 bytes	Duration in Seconds	4 bytes
	4 bytes	Timestamp of Last Sample (uint32)	e.g. 1600729945
		<i>Number of seconds since epoch (January 1, 1970 GMT).</i>	<i>(Monday, September 21, 2020 11:12:25 PM GMT)</i>
	4 bytes	SpO2 Trend Record	See Table 15: Trend Record
	4 bytes	PR Trend Record	
	4 bytes	PVi Trend Record*	
	4 bytes	RRp Trend Record*	

* This field only exists when the parameter is available per 3.4.1 Get Device Information.

Table 14: Get Trend Record Response

PAYLOAD (4 Bytes)	Size	Description	Value (s)
	1 byte	Minimum (uint8)	[0 – 255]
	1 byte	Maximum (uint8)	[0 – 255]
	1 byte	Average (uint8)	[0 – 255]
	1 byte	Last sample (uint8)	[0 – 255]

Table 15: Trend Record

3.4.5 Clear all trend

In order to clear all saved trend data for the MightySat, this **command** is sent from the Host to MightySat:

<i>PAYLOAD (1 byte)</i>	
<i>ID (1 byte)</i>	<i>Data (0 bytes)</i>
0x07	N/A

Table 16: Get Trend Record Command

Example of a command sent to MightySat: [0x77, 0x02, 0x07, 0x15]

If the command succeeds, this **ACK response** is sent from MightySat to the Host:

<i>PAYLOAD (2 bytes)</i>	
<i>ID (1 byte)</i>	<i>Data (1 byte)</i>
0xFE	0x07

If the command fails, this **NACK response** is sent from MightySat to Host:

<i>PAYLOAD (3 Bytes)</i>		
<i>ID (1 byte)</i>	<i>Data (2 bytes)</i>	
0xFF	0x07	Error Code (1 byte)